

Dr. Guangwei Gao

Curriculum Vitae

Department of Applied Mathematics
The Hong Kong Polytechnic University
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Academic Appointment

The Hong Kong Polytechnic University Postdoctoral Fellow, Department of Applied Mathematics	Aug. 2023–Present
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Education

Peking University Ph.D. in Mathematics, specialising in Computational Mathematics	Sept. 2018–Jul. 2023
Jilin University B.S. in Mathematics	Sept. 2014–Jul. 2018

Research Interests

Computational and applied mathematics, with particular interests in:

- Computing intrinsic geometric problems, including Ricci flow, isometric embedding of Riemannian manifolds, and finite element tensor calculus.
- PDEs on surfaces, free boundary problems, and geometric flows, including mean curvature flow and surface diffusion.
- Elliptic equations in non-divergence form and fully nonlinear PDEs, including Hamilton–Jacobi–Bellman and Monge–Ampère equations.
- Auxiliary space preconditioners and fast solvers for finite element discretizations.
- Structure-preserving numerical methods.
- Non-standard finite element methods.

Publications and Accepted Papers

- [1] **G. Gao**, E. S. Gawlik, B. Li. Convergence of finite element methods for Ricci flow. **SIAM Journal on Numerical Analysis**, accepted for publication. [arXiv:2607.17051](https://arxiv.org/abs/2607.17051). (Impact factor: 2.9; Journal ranking: 22/344, Q1).
- [2] **G. Gao**, B. Li, R. Tang. Convergent finite element approximations of surface evolution with relaxed minimal deformation. **Numerische Mathematik** 158 (2026), 671–714. DOI: 10.1007/s00211-026-01529-3. (Impact factor: 2.2; Journal ranking: 53/344, Q1).
- [3] **G. Gao**, H. Garcke, B. Li, R. Tang. An energy-stable minimal deformation rate scheme for mean curvature flow and surface diffusion. **SIAM Journal on Scientific Computing** 48 (2026), A103–A131. DOI: 10.1137/25M1753838. (Impact factor: 2.6; Journal ranking: 35/344, Q1).
- [4] **G. Gao**, B. Li. Geometric-structure preserving methods for surface evolution in curvature flows with minimal deformation formulations. **Journal of Computational Physics** 524 (2025), article 113718. DOI: 10.1016/j.jcp.2025.113718. (Impact factor: 3.8; Journal ranking: 2/61, Q1).
- [5] **G. Gao**, S. Wu. A C^0 finite element approximation of planar oblique derivative problems in non-divergence form. **Journal of Computational and Applied Mathematics** 427 (2023), article 115146. DOI: 10.1016/j.cam.2023.115146. (Impact factor: 2.6; Journal ranking: 35/344, Q1).
- [6] **G. Gao**, S. Wu. Auxiliary space preconditioners for a C^0 finite element approximation of Hamilton–Jacobi–Bellman equations with Cordes coefficients. **Journal of Scientific Computing** 92 (2022), article 105. DOI: 10.1007/s10915-022-01957-x. (Impact factor: 3.3; Journal ranking: 12/344, Q1).

Manuscripts under Review or Revision

- [7] **G. Gao**, B. Li, R. Tang. Dual formulations of geometric curvature flows and their discretizations. Revised manuscript submitted to **IMA Journal of Numerical Analysis**. [arXiv:2604.18288](#).
- [8] **G. Gao**, K. Hu, B. Li, G. Zhang. Finite element methods for isometric embedding of Riemannian manifolds. Under revision for **Foundations of Computational Mathematics**. [arXiv:2602.18722](#).

Teaching Experience

Teaching Assistant, Peking University

Taught tutorial classes; graded assignments and examinations.

Linear Algebra	Spring 2022
Mathematical Analysis III	Autumn 2021
Advanced Mathematics	Autumn 2019–Spring 2021
C Language	Spring 2019
Numerical Algebra	Autumn 2018

Conference Presentations

- Workshop on “Finite Element Tensor Calculus”, Tsinghua Sanya International Mathematics Forum, Sanya, China. Jan. 2026
Title: *Convergence of finite element methods for Ricci flow*.
- The 30th Biennial Numerical Analysis Conference, University of Strathclyde, Glasgow, UK. Jun. 2025
Title: *Convergence of finite element methods for Ricci flow*.
- The 8th Seminar on Numerical Methods for PDEs, Peking University, Beijing, China. Oct. 2022
Title: *A C^0 finite element approximation and fast solver of elliptic equations in non-divergence form with Cordes coefficients*.
- The 4th Peking University Workshop for Ph.D. Students in Computational and Applied Mathematics, Peking University, Beijing, China. Sept. 2021
Title: *Auxiliary space preconditioners for C^0 finite element approximation of Hamilton–Jacobi–Bellman equations with Cordes coefficients*.

Scientific Computing

Programming languages: Python, MATLAB, and C++.

Finite element software: NGSolve and FEniCS.